

Jackson Blazek

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EDUCATION

Rochester Institute of Technology (RIT)

BS/ME in Mechanical Engineering | Minors: Music Performance, Electrical Engineering

Rochester, NY

Expected May 2028

- GPA: 3.95/4.0 | RIT Honors Program
- Awards: Presidential Scholar Award, Performing Arts Scholar Award, Dean's List (2023–2025)
- Courses: Strength of Materials II, Product Innovation, Dynamics, Heat Transfer, Statics, Thermodynamics

EXPERIENCE

SpaceX

Starship Electronics Intern

Hawthorne, CA

January 2026 – Present

- Led end-to-end mechanical integration of flight-critical avionics assemblies for Starship, driving design iterations from concept to hardware release.
- Designed and iterated mechanical hardware for flight electronics assemblies in Siemens NX, optimizing vibration, thermal, and rapid-assembly constraints across high-rate production.
- Evaluated and down-selected corrosion-resistant coating systems for Starship avionics and structural hardware, informing coating specifications for flight hardware qualification.
- Executed tolerance stack-up and CTE analyses alongside DFM reviews on high-rate production hardware, validating structural and thermal compatibility of avionics assemblies and reducing integration risk ahead of flight.
- Integrated avionics hardware across Starship flight systems in collaboration with electrical, structures, and production teams, resolving compatibility requirements and implementing corrective actions to maintain integration schedule.

Avidyne Corporation

Mechanical Engineering Co-op

Melbourne, FL

May 2025 – December 2025

- Designed mechanical integration of new PCB layouts into cockpit display assemblies using SolidWorks, improving design efficiency and reducing assembly complexity.
- Executed environmental qualification testing of cockpit displays across vibration, temperature, and shock conditions, ensuring hardware reliability under DO-160G criteria.
- Validated adherence to RTCA certification standards including DO-160G, DO-186B, DO-229D, and DO-229D using DOORS to support operational compliance.

PERFORMANCE TEAMS

RIT Launch Initiative

Propulsion Engineer

Rochester, NY

September 2023 – Present

- Member of the Liquids Propulsion Team, focusing on thrust chamber and nozzle design for a 320 lbf liquid rocket engine using isopropyl alcohol and nitrous oxide.
- Conduct analyses for nozzle geometry, heat transfer, and combustion performance to ensure optimal efficiency and structural integrity at nominal conditions.
- Collaborate with the manufacturing and testing upperclassmen to develop CAD models, machine hardware, and prepare for engine hot-fire testing.
- Previously served as a Structures Engineer for Level 2 solid rockets, managing fin geometry, stability calculations, and aerodynamic performance for a 6,000-ft apogee vehicle.
- Regularly present technical progress and design justifications during team design reviews and subsystem meetings.

PROJECTS

Propeller Shaft Design for Nuclear-Powered Submarines – MECE 350 (Strengths II)

April 2025

- Designed a propeller shaft to meet torque, deflection, and fatigue constraints using MATLAB-based stress and dynamic analysis.
- Used technical torsional, critical speed, and bearing load calculations to optimize material selection and geometry.
- Crafted insightful design charts and an in-depth technical report with MATLAB, improving decision-making capabilities

Lunar Lander Simulation – MECE-117 (Introduction to Coding for Engineers)

December 2023

- Developed a MATLAB program to simulate the descent of a lunar lander under varying thrust and mass conditions.
- Modeled kinematics and fuel consumption to achieve a controlled touchdown while minimizing fuel usage.

SKILLS

Software: SolidWorks, Fusion 360, MATLAB, DOORS, Arduino, OpenRocket, Microsoft Office 365

Fabrication & Tools: Lathe, Mill, Drill Press, Bandsaw, 3D Printing, Laser Cutting, Soldering

Technical Focus Areas: CAD Modeling, Design for Manufacturing (DFM)

Hobbies: French Horn, Piano, Rowing